INCH-POUND

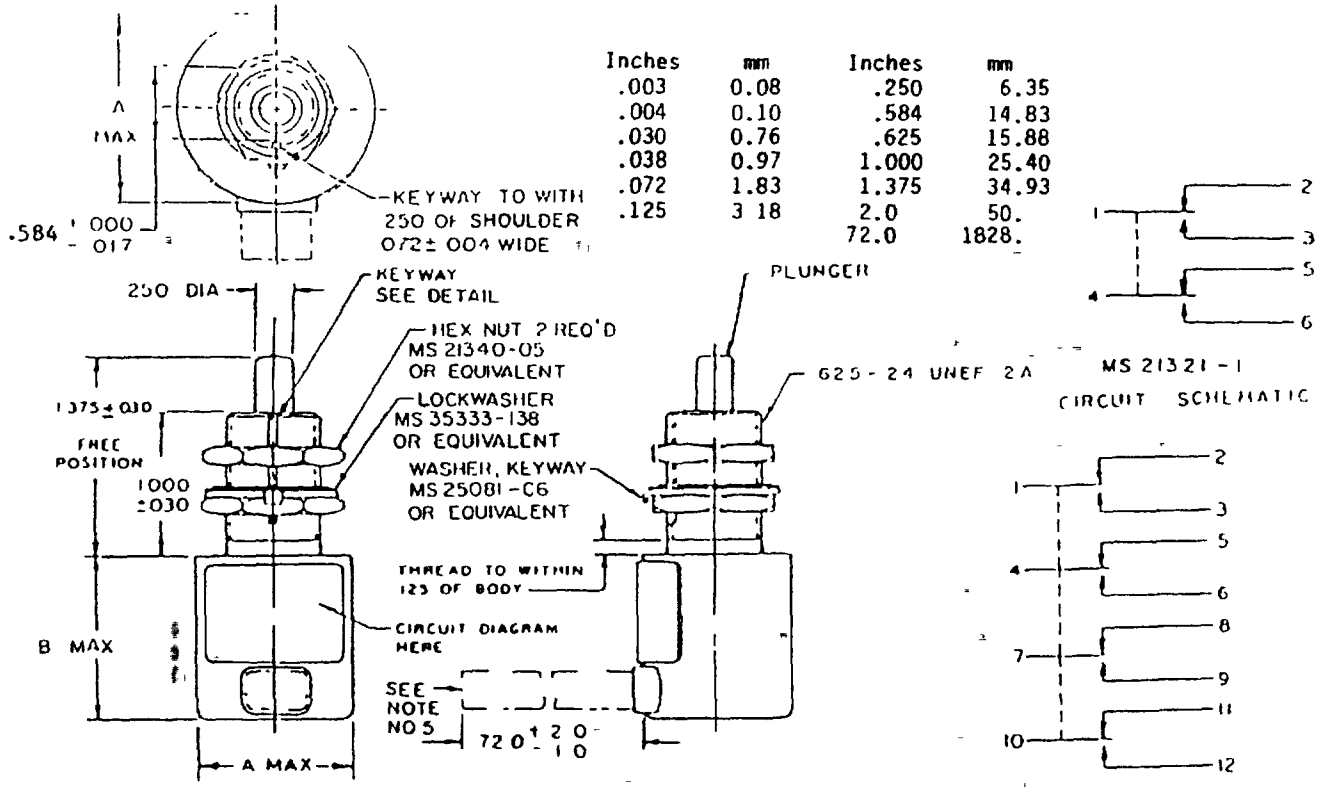
MIL-S-8805/39E
3 December 1990
SUPERSEDING
MIL-S-8805/39D
17 March 1983

MILITARY SPECIFICATION SHEET

SWITCHES, SENSITIVE, LIMIT, PLUNGER, 7 AMPERES, RESILIENT SEAL, FLUID RESISTANT

This specification is approved for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-8805.



Inches	mm	Inches	mm
.003	0.08	.250	6.35
.004	0.10	.584	14.83
.030	0.76	.625	15.88
.038	0.97	1.000	25.40
.072	1.83	1.375	34.93
.125	3 18	2.0	50.
		72.0	1828.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Contour optional, provided maximum dimensions specified are not exceeded.
5. Leads shall be M22759/7-20 wire marked in accordance with MIL-W-5088 with the switch circuit identification number followed by wire gauge number (1-20, 2-20, etc.).

FIGURE 1. Configuration and dimensions.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited

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REQUIREMENTS

Dimensions and configuration See figure 1 and table I

TABLE I. Part or Identifying Number (PIN) and characteristics

PIN	A diameter maximum	B maximum	Lead wires, number required
MS21321-1	720 (18 29)	.980 (24.89)	6
MS21321-2	1 031 (26.19)	1 200 (30 48)	12

Enclosure design: Four (resilient) All entrances to the switch cavity except through the actuator bushing shall be sealed by fusion of glass-to-metal, metal-to-metal, or ceramic-to-metal and the lead wires shall be potted to provide stress relief

Temperature characteristic One (-55°C to +85°C)

Shock type M (100 G)

Sinusoidal vibration grade: Two (10 to 2,000 Hz).

Material

Internal switches. Type CF6 in accordance with MIL-M-14 may be used.

Finish Switch housing shall be processed to resist corrosion.

Maximum weight with leads.

MS21321-1: .45 pound.

MS21321-2: .80 pound

Operating characteristics:

Actuating force 9 ±3 pounds

Overtravel force. 30 pounds maximum.

Release force 4 pounds minimum

Pretravel. .040 inch (1.01 mm) maximum.

Movement differential: 020 inch (0.50 mm) maximum.

Overtravel .250 inch (6 35 mm) minimum.

} ±20 percent variation from specified values acceptable after test

Coincidence of operating and releasing points: All poles shall transfer within .010 inch (0.25 mm) of plunger travel.

Strength of actuating means 35 pounds

Contact resistance: Not applicable.

Insulation resistance: 100 megohms

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Dielectric withstanding voltage

At atmospheric pressure 1,000 V rms

At reduced barometric pressure. 50,000 feet, 400 V rms.

Mechanical endurance. 25,000 cycles.

Electrical endurance: 25,000 cycles.

Electrical ratings See table II.

Fluid resistance Except for the cut end of the lead wire, switches shall be submerged in each of the following fluids for 2 minutes to 2 minutes 30 seconds, which shall consist of one cycle (one cycle is 10 minutes to 12 minutes 30 seconds total) Each switch shall be subjected to three cycles.

a Turbine fuel (MIL-T-5624).

b. Hydraulic fluid (SAE AS1241A).

c. Coolanol 1/ (MIL-C-47220).

d. Ethylene glycol (MIL-E-9500).

e. Lubricating oil (MIL-L-7808).

After each immersion, the excess fluid shall be blown off the external surfaces of the switch with an air jet. Following the third cycle, the switch shall be subjected to and shall meet the requirements for dielectric withstanding voltage, insulation resistance, operating characteristics, seal tests, and marking visibility.

Marking The circuit schematic shall be marked on the switch case.

PIN. See table I.

TABLE II. Electrical ratings.

Load	Sea level 28 V dc	50,000 feet 28 V dc
	(amperes)	(amperes)
Resistive	7	7
Inductive	4	2.5
Motor	4	4 1/

1/ Application information only.

Qualification inspection:

Group submission: See table III.

Group A inspection:

Seal test· Only watertight test shall be performed.

1/ Monsanto Company registered trademark.

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TABLE III Qualification inspection (group submission)

Examination or test	Samples	Extent of approval
Qualification inspection table of MIL-S-8805	MS21321-2 (24 units)	All
Visual and mechanical examination Operating characteristics	MS21321-1 (2 units)	

Revision letters are not used to denote changes due to the extensiveness of the changes

CONCLUDING MATERIAL

Custodians.
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR
Navy - AS, OS
Air Force - 99
DLA - ES

User activities
Army - AV, MI
Navy - MC
Air Force - 19

Preparing activity.
Navy - EC

Agent:
DLA - ES

(Project 5930-1413-01)

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